Lecture 1: Class Overview and Background Review

Garrett Credi

09/08/2026

Welcome to MATH 1271!



Figure: Our Mascot, Maxwell, Deep in Thought

Today's Goals

1. Overview of the Course

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2. Lecture on Algebra/Precalc Background

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1. Overview of the Course
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3. Worksheet Time!

Course Policies I

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2. We will then begin an 75 minute lecture, with a 5-ish minute break about halfway through
3. After another short break I'll hand out worksheets where we'll get a chance to apply the concepts we learned during lecture

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1. Be graded on an *honest completion* basis
2. Be split up into certain “levels”
3. And will conclude with one “challenge problem” designed to help you all develop more confidence for quiz/exam environments

Course Policies III

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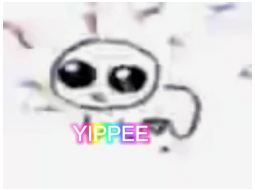
Which will all take place during our regular class meeting here (Tate Hall 120).
Finally we will have a *cumulative* final exam on 12/17 from 12PM-3PM Location TBD.

Course Policies V

Some things that might make you happy!

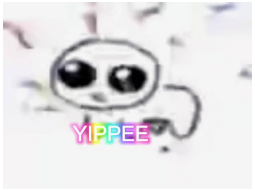
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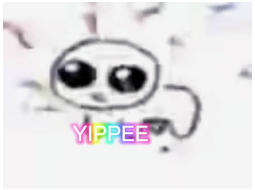
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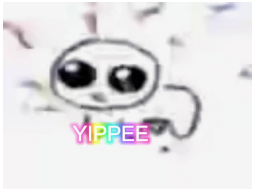
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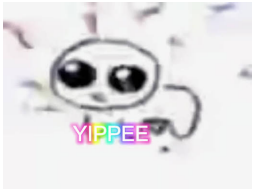
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Office Hours?

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I'm planning on holding two hours of office hours a week: in order that I hold them at the most opportune times please indicate your desires/availability on the When2Meet here.
There are also Office Hours that are held by some other TAs that you can find here.

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You can contact me at credi008@umn.edu

Who are you?

What is your:

1. Name

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1. Name
2. Major and standing (freshman, sophomore, etc)
3. Favorite *thing*?

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3. Trigonometric functions
4. Solving Equations

Functions I

What is a function?

Function

Equation of a Function

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A function is an association from “things” of type A to other “things” of type B, typically denoted as $f : A \to B$.

Equation of a Function

Real valued functions of a single real variable (we’ll just call these functions from now on) can be **defined** by an equation

E.g. we can define the *squaring* function as $f(x) = x^2$.

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E.g. we can define the *squaring* function as $f(x) = x^2$. This means the value of the function f at some real number x is the square of x, so, for instance, $f(2) = 2^2 = 4, f(3) = 3^2 = 9$, etc.

Functions II

Graph of a function

The **graph** of a function $f : \mathbb{R} \to \mathbb{R}$

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The **graph** of a function $f : \mathbb{R} \to \mathbb{R}$ is a graphical/geometric way of interpreting what a function does.
To draw the graph, we first draw a *2D-Coordinate Grid*, and then draw all the points $(x, f(x))$ connected via a smooth line.

E.g.

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A function is **defined piecewise** by the following method:

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$$wacky(x) = \begin{cases} x^2, & x > -5 \\ -x^3 + 2x + 1, & -5 \leq x < 2 \\ 1, & 2 \leq x < 3 \\ \frac{x}{2} + 3, & \text{otherwise} \end{cases} \tag{3}$$

Functions V

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What about a function like $f(x) = x^2 + 1$?

Fractions I

It's also worth going over some fraction foundations which will be very useful very soon in the course:

Addition of Fractions

Multiplication of Fractions

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Fractions II

I also want to make clear what people mean by:

The Multiplying by 1 Trick

Fractions II

I also want to make clear what people mean by:

- Clearing out common factors/
- Factoring [blah] out/
- Multiplying by 1

The Multiplying by 1 Trick

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- The exponential function $\exp(x) = e^x$
- It’s inverse, the natural logarithm $\ln(x)$
- The power functions x^n
- Their inverses $\sqrt[n]{x}$
- The trigonometric functions

Trigonometric Functions

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For instance, let's say we want to buy some ramen and the cost of ramen in dollars varies by weight in pounds according to the following function:

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If we only have 30 dollars, how much ramen can we buy?

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So we can use our definition of P to get:

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$$w^2 + 7w - 30 = 0$$
$$(w + 10)(w - 3) = 0$$

Solving Equations II

How do we set that up mathematically?
We want to find the weight w so that:

$$P(w) = \$30 \tag{4}$$

So we can use our definition of P to get:

$$\begin{aligned} w^2 + 7w &= 30 \\ w^2 + 7w - 30 &= 0 \\ (w + 10)(w - 3) &= 0 \\ w = -10 \text{ or } w &= 3 \end{aligned}$$

And since -10 lbs of ramen doesn't make physical sense, we get that we can by 3 lbs of ramen with that much money.

Solving Equations III

Why did we choose to solve

$$(w + 10)(w - 3) = 0$$

for $w = -10, w = 3$
instead of solving

$$w(w + 7) = 30$$

for $w = 30$ or $w + 7 = 30$?

No Zero Divisors

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Solving Equations IV

How might we approach solving for x in an equation like:

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Or maybe like:

$$2\sin(x)^2 + \cos(x)^2 = 2? \tag{6}$$

Worksheet Time

Let’s get into groups and work on the worksheet!

End Of Class

That’s all folks!